

IDTA 02079-1-0

Safety Instrumented System (SIS) Device for the Process Industries

September 2026

SPECIFICATION

Submodel Template of the
Asset Administration Shell



Submodel Template

IDTA approved

- 100% AAS compliant
- Consistent & interoperable
- Released by the AAS experts

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Chapter 1. General

1.1. About this document

This document is a part of a specification series. Each part specifies the contents of a Submodel template for the Asset Administration Shell (AAS). The AAS is described in [1], [2], [3] and [6]. First exemplary Submodel contents were described in [4], while the actual format of this document was derived by the "Administration Shell in Practice" [5]. The format aims to be very concise, giving only minimal necessary information for applying a Submodel template, while leaving deeper descriptions and specification of concepts, structures and mapping to the respective documents [1] to [6]. Common terms and abbreviations can be found in [7] to [11].

The target audience of the specification are developers and editors of technical documentation and manufacturer information, which are describing assets in smart manufacturing by means of the Asset Administration Shell (AAS) and therefore need to create a Submodel instance with a hierarchy of SubmodelElements. In addition to these people working at vendors or Engineering companies, it is interesting for people at owner operators working within the functional safety lifecycle. This document especially details on the question, which SubmodelElements with which semantic identification shall be used for this purpose.

This SMT will only be fully supported as of metamodel V3.0.

1.2. Scope of the Submodel

Safety is commonly defined as the absence of unacceptable risk, and risk reduction is achieved by implementing safety barriers that prevent accidents. Functional safety ensures that instrumented barriers operate correctly when needed, making it a critical element in accident prevention. IEC 61511 is an international standard that governs the use of Safety Instrumented Systems (SIS) in the process industry, such as chemical, pharmaceutical, and energy sectors. It outlines requirements for managing the entire safety lifecycle—from initial risk assessment and system design to operation, maintenance, and eventual decommissioning. A key concept in the standard is the use of Safety Integrity Levels (SILs), which define the required performance of safety functions to reduce risks to acceptable levels. The standard also emphasizes the importance of a structured management system, including clear responsibilities, documentation, and periodic reviews to ensure ongoing safety. Based on the more general IEC 61508, IEC 61511 adapts its principles specifically for the needs and challenges of the process industry.

There is an increasing number of digitalization initiatives related to SIS, motivated by the opportunity to reduce resources while enhancing the quality of engineering and operational follow-up of SIS. Digitalization relies on data exchange in an automated and autonomous way. Mastering interoperability is crucial to fully leverage the benefits of increasing digitalization. Interoperability is a key prerequisite for successful digitalization. It requires standardizing information models and the meaning of the terms being exchanged so they are understood by people, and adding machine-readable codes so they are equally understood by machines.

The submodel SIS Device is one out of four submodels anchored in IEC 61511. The purpose of the submodel is to define and structure relevant properties derived from the safety requirements specification (SRS) allocated to the SIS device that are part of an implemented safety instrumented function (SIF), in addition to include other relevant properties for functional safety throughout the lifecycle for a SIS device, such as technical data and operational data. The submodel should be introduced from the stage when information about the functional and safety integrity level requirements becomes available. The submodel is updated with design data during the preliminary and detailed design phase, and operational data is added during operation. Updates must comply with IEC 61511's requirements for management of change. To model the dependencies between the asset Safety Instrumented System and the asset SIS device, the submodel "Hierarchical Structures enabling Bills of Material" can be used.

1.3. Relevant standards for the Submodel template

IEC 61511-1:2016+A1:2017: Functional safety - Safety instrumented systems for the process industry sector - Part 1: Framework, definitions, system, hardware and application programming requirements [7]

NE 192:2025: NAMUR Recommendation 192 - Functional Safety Information Model [8]

APOS Project Report 2023:00108: Guidelines for standardised classification and failure reporting for safety equipment in the petroleum industry [9]

APOS Project Report 2023:00107: Guideline for follow-up of Safety Instrumented Systems (SIS) in the operating phase [10]

1.4. Steps for engineering and operation of Safety Instrumented Systems

If a risk analysis shows that additional safety measures are needed to reach a defined level of safety, a SIF might be required [1] to prevent the process from entering unsafe states. Following the hazard and risk assessment, the first two steps toward developing a SIF are the allocation of protection layers and the preparation of the Safety Requirement Specification for the SIF (dark blue in [\[Figure_1\]](#)) and the Safety Instrumented Systems (SIS) that realize the SIF.

The SIS part (green in [\[Figure_1\]](#)) involves the design and engineering of the SIS and its SIS devices. At this stage, the instrumentation and control engineers specify all SIS devices needed to implement the required safety functions. This includes input devices, final elements, logic solvers, infrastructure devices and electrical components.

The implementation of hardware and software is then validated and commissioned. After commissioning, it becomes the asset in operation (pink in [\[Figure_1\]](#)), representing the Operation and Maintenance phase.

NOTE

figure [Figure_1](#).

Chapter 2. Information set for Submodel “Safety Instrumented System (SIS) Device for the Process Industries”

2.1. General

While defining Submodels the following three aspects must be considered as suggested in [5]:

Use and economic relevance

Quantifying the cost savings from accidents that should never occur is inherently difficult. However, it is clear that for both emerging sectors like hydrogen and established industries such as chemicals, energy, and pharmaceuticals, accident prevention is essential for public trust and acceptance. Moreover, operational disruptions and unplanned failures can be costly and may compromise the reliability of supply chains, including energy and raw materials.

Value creation potential

The following possible value creation potential that the innovation will have for the different stakeholders are identified:

Operator companies

- Improved safety and productivity – better control of major accidents and increased operational regularity due to enhanced follow-up enabled by interoperability, interoperable electronic industry formats for main safety documentation, all design data and safety requirements readily available in operation, reduced needs for manual updates, improved systems and products based on (real time) feedback from operation
- Increased productivity – reduced purchasing, and operational cost (independent of specific manufacturer solutions, scalability, and flexibility (independent of proprietary solution)

Consultants and engineering companies

- Increased productivity and improved quality of service – interoperable electronic industry formats for main safety documentation, reduced needs for manual updates, document revisions and changes during design
- Increased productivity and sales – reuse of existing models and systems which can be sold internationally and to other industry sectors
- Increased competitive edge – knowledge and competence about I4.0, digital twins (AAS) and associated human centric solutions
- New markets – better integration of matured concepts from the established industries to emerging sectors

Vendors and manufacturers

- Increased sales – improved systems and products based on (real time) data from operation, new business opportunities/business models
- Increased competitive edge – faster and more cost-effective development of new solutions, competence about I4.0, digital twins (AAS) and associated human centric solutions
- New markets – better integration of matured concepts from the established industries to emerging sectors

Possible functions and interactions

The use cases for this submodel include:

- Digitalizing SRS data for seamless exchange between manufacturers, system integrators and end users, replacing the current approach of sharing SRS data in less suitable formats like pdf, word, and excel. The SRS data are used into various types of software tools for e.g., consistency checks with referenced design documents and risk analyses, compliance checks with technical solutions, and as benchmark for performance monitoring in operation
- Digitalization of SIF design data for seamless exchange between the same listed actors, replacing documents like safety manuals in less suitable formats. SIF design data are stored in several systems, like databases for equipment listing and maintenance management
- Digital twin implementation for SIFs in operation to collect information about demands and proof test for a systematic evaluation of the operational reliability of the SIS as well as for a review of the assumptions in the design phase

Property specification

See chapter 3 "[Information structures and attributes](#)".

2.2. Overview UML model

The information model discussed in this chapter focuses on a Safety Instrumented System (SIS) Device as a typical asset. This is based on work performed in the APOS project [12]-[14] as well as results from the NAMUR Working Group 1.3 [15].

The SubmodelElements described in section 3 are structured in the following way (see [\[UML_overview_2_2\]](#)):

NOTE	UML diagram for 0 .
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The Submodel template and its instances consists of several Submodel Element Collections:

- **TagInformation** This SMC contains general information about the TAG (role) that the SIS device represents
- **SRSRequirements** This SMC contains the requirements from the Safety Requirement Specification allocated to the SIS device
- **DeviceModelSpecification** This SMC contains model (type) properties of the SIS device
- **ReliabilityDataDesign** This SMS contains reliability data of the SIS device given by the vendor or from other sources
- **ReliabilityInfluencingProperties** This SMC contains properties which are identified to have impact on the reliability of the device
- **DeviceInstanceSpecification** This SMC contains instance specifications about the SIS device
- **BypassData** This SMC contains information about bypasses of the SIS device during operation
- **Diagnostics** This SMC contains properties for diagnostic information during operation
- **DeviceOperationData** This SMC contains information about the operation time of the SIS device
- **DemandData** This SMC contains information about demands during operation related to the SIS device
- **FailureData** This SMC contains information about failures of the SIS device during operation
- **ReliabilityDataOperational** This SMS contains reliability of the SIS device monitored by the owner or operator
- **ProofTestData** This SMC contains information about proof tests performed of the SIS device

Chapter 3. Information structures and attributes

3.1. Properties of the SMC TagInformation

[UML_Submodel_3_1] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [TagInformation](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	TagInformation
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/TagInformation/1
Parent:	-
Explanation:	gives relevant information about the tag that the device represents
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TagName	https://admin-shell.io/idta/cds/TagName/1 gives information about the tag that the device represents	[xs:string]	One
[Prop] TagDescription	https://admin-shell.io/idta/cds/TagDescription/1 gives relevant information about the tag that the device represents	[xs:string]	ZeroToOne
[Prop] Subsystem	https://admin-shell.io/idta/cds/Subsystem/1 gives relevant information about the tag that the device represents	[xs:string]	One
[Prop] SafeStateOfDevice	https://admin-shell.io/idta/cds/SafeStateOfDevice/1 gives relevant information about the tag that the device represents	[xs:string]	ZeroToOne
[Prop] EquipmentType	https://admin-shell.io/idta/cds/EquipmentType/1 gives relevant information about the tag that the device represents	[xs:string]	ZeroToOne
[Prop] ReferenceToTagGroup	https://admin-shell.io/idta/cds/ReferenceToTagGroup/1 gives relevant information about the tag that the device represents	[xs:string]	ZeroToMany

3.2. Properties of the SMC SRSRequirements

[UML_Submodel_3_2] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [SRSRequirements](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	SRSRequirements
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/SRSRequirements/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] SystematicCapabilityRequirement	https://admin-shell.io/idta/cds/SystematicCapabilityRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Prop] PFDBudget	https://admin-shell.io/idta/cds/PFDBudget/1 SRS requirement(s) according to IEC 61511	[xs:float]	ZeroToOne
[Prop] PFHBudget	https://admin-shell.io/idta/cds/PFHBudget/1 SRS requirement(s) according to IEC 61511	[xs:float]	ZeroToOne
[Prop] InternalHFTRequirement	https://admin-shell.io/idta/cds/InternalHFTRequirement/1 gives information about the model of the device	[xs:string]	ZeroToOne
[SMC] ProofTest	https://admin-shell.io/idta/cds/ProofTest/1 SRS requirement(s) according to IEC 61511	[]	One
[SMC] PartialStrokeTest	https://admin-shell.io/idta/cds/PartialStrokeTest/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne
[Prop] MaximumAllowableSISDeviceResponseTime	https://admin-shell.io/idta/cds/MaximumAllowableSISDeviceResponseTime/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MinimumAllowableSISDeviceResponseTime	https://admin-shell.io/idta/cds/MinimumAllowableSISDeviceResponseTime/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	ZeroToOne
[Prop] MaximumPermittedRepairTime	https://admin-shell.io/idta/cds/MaximumPermittedRepairTime/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One
[SMC] BypassRequirements	https://admin-shell.io/idta/cds/BypassRequirements/1 SRS requirement(s) according to IEC 61511	[]	One
[Prop] TripEnergyMode	https://admin-shell.io/idta/cds/TripEnergyMode/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Prop] RequirementIfEnergizedToTrip	https://admin-shell.io/idta/cds/RequirementIfEnergizedToTrip/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] TripAction	https://admin-shell.io/idta/cds/TripAction/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Prop] SurvivabilityRequirement	https://admin-shell.io/idta/cds/SurvivabilityRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] DiagnosticRequired	https://admin-shell.io/idta/cds/DiagnosticRequired/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] AdditionalDiagnosticRequirementForImplementation	https://admin-shell.io/idta/cds/AdditionalDiagnosticRequirementForImplementation/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] TimeDelayOfAction	https://admin-shell.io/idta/cds/TimeDelayOfAction/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	ZeroToMany
[SMC] EnvironmentalExtremes	https://admin-shell.io/idta/cds/EnvironmentalExtremes/1 environmental extreme value	[]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] FailureModeResponses	https://admin-shell.io/idta/cds/FailureModeResponses/1 SRS requirement(s) according to IEC 61511	[]	ZeroToMany
[SMC] InputDevice	https://admin-shell.io/idta/cds/InputDevice/1 gives information about the input device subsystem of the SIF	[]	ZeroToOne
[SMC] LogicSolver	https://admin-shell.io/idta/cds/LogicSolver/1 gives information about the logic solver subsystem of the SIF	[]	ZeroToOne
[SMC] FinalElement	https://admin-shell.io/idta/cds/FinalElement/1 gives information about the final element subsystem of the SIF	[]	ZeroToOne
[Prop] SharedUselsAllowed	https://admin-shell.io/idta/cds/SharedUselsAllowed/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	One
[Ref] ReferenceToSIF	https://admin-shell.io/idta/cds/ReferenceToSIF/1 SRS requirement(s) according to IEC 61511	[]	One
[Prop] UsefulLife	https://admin-shell.io/idta/cds/UsefulLife/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne

3.3. Properties of the SMC DeviceModelSpecification

[UML_Submodel_3_3] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [DeviceModelSpecification](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	DeviceModelSpecification
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/DeviceModelSpecification/1
Parent:	-
Explanation:	gives information about the model of the device
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[File] Instrument DataSheet	https://admin-shell.io/idta/cds/InstrumentDataSheet/1 gives information about the model of the device	[]	One
[File] SafetyManual	https://admin-shell.io/idta/cds/SafetyManual/1 gives information about the model of the device	[]	One
[Prop] ProductType peName	https://admin-shell.io/idta/cds/ProductTypeName/1 gives information about the model of the device	[xs:string]	One
[Prop] ProductFamily Name	https://admin-shell.io/idta/cds/ProductFamilyName/1 gives information about the model of the device	[xs:string]	ZeroToOne
[Prop] ProductModel Name	https://admin-shell.io/idta/cds/ProductModelName/1 gives information about the model of the device	[xs:string]	ZeroToOne
[Prop] NameOfManufacturer	https://admin-shell.io/idta/cds/NameOfManufacturer/1 gives information about the model of the device	[xs:string]	One
[Prop] DeviceDescription	https://admin-shell.io/idta/cds/DeviceDescription/1 gives information about the model of the device	[xs:string]	ZeroToOne
[Prop] DeviceBoundary	https://admin-shell.io/idta/cds/DeviceBoundary/1 gives information about the model of the device	[xs:string]	ZeroToOne
[Prop] MaintainableItem	https://admin-shell.io/idta/cds/MaintainableItem/1 gives information about the model of the device	[xs:string]	ZeroToMany
[Prop] SystematicCapability Specified	https://admin-shell.io/idta/cds/SystematicCapabilitySpecified/1 gives information about the model of the device	[xs:string]	One
[Prop] InternalHFTSpecified	https://admin-shell.io/idta/cds/InternalHFTSpecified/1 gives information about the model of the device	[xs:string]	ZeroToOne
[Prop] IsPriorUse	https://admin-shell.io/idta/cds/IsPriorUse/1 gives information about the model of the device	[xs:boolean]	One
[Prop] IsProvenInUse	https://admin-shell.io/idta/cds/IsProvenInUse/1 gives information about the model of the device	[xs:boolean]	One
[Prop] MeanTimeToRestoration	https://admin-shell.io/idta/cds/MeanTimeToRestoration/1 gives information about the model of the device	[xs:double]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] FinalElement	https://admin-shell.io/idta/cds/FinalElement/1 gives information about the final element subsystem of the SIF	[]	ZeroToOne
[SMC] LogicSolver	https://admin-shell.io/idta/cds/LogicSolver/1 gives information about the logic solver subsystem of the SIF	[]	ZeroToOne
[SMC] InputDevice	https://admin-shell.io/idta/cds/InputDevice/1 gives information about the input device subsystem of the SIF	[]	ZeroToOne
[Prop] TypeAOrB	https://admin-shell.io/idta/cds/TypeAOrB/1 gives information about the model of the device	[xs:string]	ZeroToOne
[Prop] HardwareVersion	https://admin-shell.io/idta/cds/HardwareVersion/1 gives information about the instance of the device	[xs:string]	ZeroToOne
[Prop] SoftwareVersion	https://admin-shell.io/idta/cds/SoftwareVersion/1 gives information about the instance of the device	[xs:string]	ZeroToOne

3.4. Properties of the SMC ReliabilityDataDesign

[UML_Submodel_3_4] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [ReliabilityDataDesign](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	ReliabilityDataDesign
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ReliabilityDataDesign/1
Parent:	-
Explanation:	reliability data parameter(s) with design data value(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MeanTimeBetweenFailures	https://admin-shell.io/idta/cds/MeanTimeBetweenFailures/1 reliability data parameter(s) with design data value(s)	[xs:decimal]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] lambdaDU Design	https://admin-shell.io/idta/cds/LambdaDUDesign/1 reliability data parameter(s) with design data value(s)	[xs:double]	One
[Prop] lambdaDD Design	https://admin-shell.io/idta/cds/LambdaDDDesign/1 reliability data parameter(s) with design data value(s)	[xs:double]	ZeroToOne
[Prop] lambdaSU Design	https://admin-shell.io/idta/cds/LambdaSUDesign/1 reliability data parameter(s) with design data value(s)	[xs:double]	ZeroToOne
[Prop] lambdaSD Design	https://admin-shell.io/idta/cds/LambdaSDDesign/1 reliability data parameter(s) with design data value(s)	[xs:double]	ZeroToOne
[Prop] Beta	https://admin-shell.io/idta/cds/Beta/1 reliability data parameter(s) with design data value(s)	[xs:decimal]	ZeroToOne
[Prop] SafeFailureFraction Design	https://admin-shell.io/idta/cds/SafeFailureFractionDesign/1 reliability data parameter(s) with design data value(s)	[xs:decimal]	One
[Prop] Diagnostic Coverage Design	https://admin-shell.io/idta/cds/DiagnosticCoverageDesign/1 reliability data parameter(s) with design data value(s)	[xs:decimal]	One
[Prop] DataSource	https://admin-shell.io/idta/cds/DataSource/1 source of the reliability data in this SMC	[xs:string]	OneToMany

3.5. Properties of the SMC

ReliabilityInfluencingProperties

[UML_Submodel_3_5] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [ReliabilityInfluencingProperties](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	ReliabilityInfluencingProperties
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ReliabilityInfluencingProperties/1
Parent:	-
Explanation:	reliability influencing property for the device

Element details:	
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[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Application	https://admin-shell.io/idta/cds/Application/1 reliability influencing property for the device	[xs:string]	ZeroToMany
[Prop] ExternalExposure	https://admin-shell.io/idta/cds/ExternalExposure/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] FluidSeverity	https://admin-shell.io/idta/cds/FluidSeverity/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] Fluid	https://admin-shell.io/idta/cds/Fluid/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[SMC] InputDevice	https://admin-shell.io/idta/cds/InputDevice/1 gives information about the input device subsystem of the SIF	[]	ZeroToOne
[SMC] LogicSolver	https://admin-shell.io/idta/cds/LogicSolver/1 gives information about the logic solver subsystem of the SIF	[]	ZeroToOne
[SMC] FinalElement	https://admin-shell.io/idta/cds/FinalElement/1 gives information about the final element subsystem of the SIF	[]	ZeroToOne
[Prop] InternalDiagnostic	https://admin-shell.io/idta/cds/InternalDiagnostic/1 reliability influencing property for the device	[xs:string]	ZeroToMany
[Prop] ExternalDiagnostic	https://admin-shell.io/idta/cds/ExternalDiagnostic/1 reliability influencing property for the device	[xs:string]	ZeroToMany

3.6. Properties of the SMC DeviceInstanceSpecification

[UML_Submodel_3_6] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [DeviceInstanceSpecification](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	DeviceInstanceSpecification
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/DeviceInstanceSpecification/1
Parent:	-

Explanation:	gives information about the instance of the device
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Equipment Number	https://admin-shell.io/idta/cds/EquipmentNumber/1 gives information about the instance of the device	[xs:string]	ZeroToOne
[Prop] SerialNum ber	https://admin-shell.io/idta/cds/SerialNumber/1 gives information about the instance of the device	[xs:string]	One
[Prop] DateOfPro duction	https://admin-shell.io/idta/cds/DateOfProduction/1 gives information about the instance of the device	[xs:string]	One
[Prop] StartDate Operation	https://admin-shell.io/idta/cds/StartDateOperation/1 gives information about the instance of the device	[xs:date]	ZeroToOne
[Prop] EndDateO peration	https://admin-shell.io/idta/cds/EndDateOperation/1 gives information about the instance of the device	[xs:date]	ZeroToOne
[Prop] PhysicalLo cation	https://admin-shell.io/idta/cds/PhysicalLocation/1 gives information about the tag that the device represents	[xs:string]	ZeroToOne
[Ref] Reference ToAssocia tedDevice	https://admin-shell.io/idta/cds/ReferenceToAssociatedDevice/1 gives information about the instance of the device	[]	ZeroToMa ny
[Ref] Reference ToModelN umber	https://admin-shell.io/idta/cds/ReferenceToModelNumber/1 gives relevant information about the instance that the device represents	[]	One
[Ref] Reference ToTag	https://admin-shell.io/idta/cds/ReferenceToTag/1 gives information about the instance of the device	[]	One

3.7. Properties of the SMC BypassData

[UML_Submodel_3_7] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [BypassData](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	BypassData
----------	------------

Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/BypassData/1
Parent:	-
Explanation:	gives information about occurred bypass(es)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] NumberOfBypasses	https://admin-shell.io/idta/cds/NumberOfBypasses/1 gives information about occurred bypass(es)	[xs:integer]	One
[Prop] TotalBypassDuration	https://admin-shell.io/idta/cds/TotalBypassDuration/1 gives information about occurred bypass(es)	[xs:integer]	One
[SMC] BypassEvent	https://admin-shell.io/idta/cds/BypassEvent/1 gives information about occurred bypass(es)	[]	ZeroToMany

3.8. Properties of the SMC Diagnostics

[UML_Submodel_3_8] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for **Diagnostics**.

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	Diagnostics
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/Diagnostics/1
Parent:	-
Explanation:	gives information about the device diagnostics
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] DiagnosticFunction	https://admin-shell.io/idta/cds/DiagnosticFunction/1 gives information about the device diagnostics	[xs:string]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Rel] Reference ToDiagnosticAlarm	https://admin-shell.io/idta/cds/ReferenceToDiagnosticAlarm/1 gives information about the device diagnostics	[]	ZeroToMany
[SMC] AlarmEventData	https://admin-shell.io/idta/cds/AlarmEventData/1 gives information about the device diagnostics	[]	ZeroToMany

3.9. Properties of the SMC DeviceOperationData

[UML_Submodel_3_9] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [DeviceOperationData](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	DeviceOperationData
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/DeviceOperationData/1
Parent:	-
Explanation:	reliability data parameter(s) with operational data value(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Operating Time	https://admin-shell.io/idta/cds/OperatingTime/1 operational data of the device	[xs:double]	One
[Prop] SurveillanceTime	https://admin-shell.io/idta/cds/SurveillanceTime/1 operational data of the device	[xs:double]	One
[Prop] DeviceOperationalStatus	https://admin-shell.io/idta/cds/DeviceOperationalStatus/1 operational data of the device	[xs:string]	One
[SMC] ConditionMonitoringData	https://admin-shell.io/idta/cds/ConditionMonitoringData/1 operational data of the device	[]	ZeroToMany

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] FailureStatusSignal	https://admin-shell.io/idta/cds/FailureStatusSignal/1 operational data of the device	[xs:string]	One
[Prop] PresentTestInterval	https://admin-shell.io/idta/cds/PresentTestInterval/1 gives information about test(s)	[xs:string]	ZeroToOne
[Prop] ActualMeanRepairTime	https://admin-shell.io/idta/cds/ActualMeanRepairTime/1 gives information about occurred failure(s)	[xs:double]	One
[Prop] MissionTime	https://admin-shell.io/idta/cds/MissionTime/1 operational data of the device	[xs:string]	ZeroToOne

3.10. Properties of the SMC DemandData

[UML_Submodel_3_10] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [DemandData](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	DemandData
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/DemandData/1
Parent:	-
Explanation:	gives information about occurred demand(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] OperationalDemandRate	https://admin-shell.io/idta/cds/OperationalDemandRate/1 gives information about occurred demand(s)	[xs:decimal]	One
[Prop] NumberOfDemands	https://admin-shell.io/idta/cds/NumberOfDemands/1 gives information about occurred demand(s)	[xs:integer]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] DeviceDemandEvent	https://admin-shell.io/idta/cds/DeviceDemandEvent/1 gives information about occurred demand(s)	[]	ZeroToMany

3.11. Properties of the SMC FailureData

[UML_Submodel_3_11] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for **FailureData**.

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	FailureData
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FailureData/1
Parent:	-
Explanation:	gives relevant information about the occurred failure(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] NumberOfFailures	https://admin-shell.io/idta/cds/NumberOfFailures/1 gives information about occurred failure(s)	[xs:integer]	One
[Prop] NumberOfDUFailures	https://admin-shell.io/idta/cds/NumberOfDUFailures/1 gives information about occurred failure(s)	[xs:integer]	One
[Prop] NumberOfSpuriousTrips	https://admin-shell.io/idta/cds/NumberOfSpuriousTrips/1 gives information about occurred failure(s)	[xs:integer]	One
[SMC] FailureEvent	https://admin-shell.io/idta/cds/FailureEvent/1 gives information about occurred failure(s)	[]	ZeroToMany

3.12. Properties of the SMC ReliabilityDataOperational

[UML_Submodel_3_12] shows the UML-diagram defining the relevant properties which need to be set.

NOTEUML diagram for [ReliabilityDataOperational](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	ReliabilityDataOperational
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ReliabilityDataOperational/1
Parent:	-
Explanation:	reliability data parameter(s) with operational data value(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] lambdaDU Operational	https://admin-shell.io/idta/cds/LambdaDUOperational/1 reliability data parameter(s) with operational data value(s)	[xs:double]	One
[Prop] PFHAverageBasedOnOperationalData	https://admin-shell.io/idta/cds/PFHAverageBasedOnOperationalData/1 performance measure(s)	[xs:double]	ZeroToOne
[Prop] PFDAverageBasedOnOperationalData	https://admin-shell.io/idta/cds/PFDAverageBasedOnOperationalData/1 reliability data parameter(s) with operational data value(s)	[xs:decimal]	ZeroToOne
[Prop] lambdaDD Operational	https://admin-shell.io/idta/cds/LambdaDDOperational/1 reliability data parameter(s) with operational data value(s)	[xs:double]	ZeroToOne
[Prop] lambdaSU Operational	https://admin-shell.io/idta/cds/LambdaSUOperational/1 reliability data parameter(s) with operational data value(s)	[xs:double]	ZeroToOne
[Prop] SpuriousTripRateOperational	https://admin-shell.io/idta/cds/SpuriousTripRateOperational/1 gives information about occurred demand(s)	[xs:double]	One
[Prop] lambdaSD Operational	https://admin-shell.io/idta/SISDevice/LambdaSDOperational/1/0 reliability data parameter(s) with operational data value(s)	[xs:double]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] SafeFailureFractionOperational	https://admin-shell.io/idta/cds/SafeFailureFractionOperational/1 reliability data parameter(s) with operational data value(s)	[xs:decimal]	ZeroToOne
[Prop] DiagnosticCoverageOperational	https://admin-shell.io/idta/cds/DiagnosticCoverageOperational/1 reliability data parameter(s) with operational data value(s)	[xs:decimal]	ZeroToOne

3.13. Properties of the SMC ProofTestData

[UML_Submodel_3_13] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for [ProofTestData](#).

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	ProofTestData
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ProofTestData/1
Parent:	-
Explanation:	gives information about test(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TypeOfTest	https://admin-shell.io/idta/cds/TypeOfTest/1 gives information about test(s)	[xs:string]	One
[Prop] TestDescription	https://admin-shell.io/idta/cds/TestDescription/1 gives information about test(s)	[xs:string]	ZeroToOne
[Prop] ShutdownIsNeededForTesting	https://admin-shell.io/idta/cds/ShutdownIsNeededForTesting/1 gives information about test(s)	[xs:boolean]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MinimumTimeRequiredForTesting	https://admin-shell.io/idta/cds/MinimumTimeRequiredForTesting/1 gives information about test(s)	[xs:double]	ZeroToOne
[Prop] RequiredResources	https://admin-shell.io/idta/cds/RequiredResources/1 gives information about test(s)	[xs:string]	ZeroToOne
[Prop] PresentTestInterval	https://admin-shell.io/idta/cds/PresentTestInterval/1 gives information about test(s)	[xs:string]	One
[Prop] SourceOfPresentTestInterval	https://admin-shell.io/idta/cds/SourceOfPresentTestInterval/1 gives information about test(s)	[xs:string]	OneToMany
[Prop] TestDuration	https://admin-shell.io/idta/cds/TestDuration/1 gives information about test(s)	[xs:double]	ZeroToOne
[Prop] NumberOfPerformedTests	https://admin-shell.io/idta/cds/NumberOfPerformedTests/1 gives information about test(s)	[xs:string]	One
[Prop] NumberOfFailedTests	https://admin-shell.io/idta/cds/NumberOfFailedTests/1 gives information about test(s)	[xs:string]	One
[Ref] ReferenceToFailCriterion	https://admin-shell.io/idta/cds/ReferenceToFailCriterion/1 gives information about test(s)	[]	ZeroToOne
[SMC] ProofTestEvent	https://admin-shell.io/idta/cds/ProofTestEvent/1 gives information about test(s)	[]	ZeroToMany

Annex A. Explanations on used table formats

1. General

The used tables in this document try to outline information as concise as possible. They do not convey all information on Submodels and SubmodelElements. For this purpose, the definitive definitions are given by a separate file in form of an AASX file of the Submodel template and its elements.

2. Tables on Submodels and SubmodelElements

For clarity and brevity, a set of rules is used for the tables for describing Submodels and SubmodelElements.

- The tables follow in principle the same conventions as in [5].
- The table heads abbreviate 'cardinality' with 'card'.
- The tables often place two informations in different rows of the same table cell. In this case, the first information is marked out by sharp brackets [] from the second information. A special case are the semanticIds, which are marked out by the format: (type)(local)[idType]value.
- The types of SubmodelElements are abbreviated (see Table 1):

Table 1. Abbreviations for SubmodelElements

SME type	SubmodelElement type
Blob	Blob
Cap	Capability
Ent	Entity
Evt	Event
File	File
MLP	MultiLanguageProperty
Opr	Operation
Prop	Property
Range	Range
Ref	ReferenceElement
Rel	RelationshipElement
RelA	AnnotatedRelationshipElement
SMC	SubmodelElementCollection
SME	SubmodelElement type
SML	SubmodelElementList

- If an idShort ends with '__00__', this indicates a suffix of the respective length (here: 2) of decimal digits, in order to make the idShort unique. A different idShort might be chosen, as long as it is unique in the parent's context.
- The Keys of semanticId in the main section feature only idType and value, such as: <https://admin-shell.io/vdi/2770/1/0/DocumentId/Id>. The attribute "type" (typically "ConceptDescription" and "(local)" or

"GlobalReference") need to be set accordingly; see [6].

- If a table does not contain a column with "parent" heading, all represented attributes share the same parent. This parent is denoted in the head of the table.
- Multi-language strings are represented by the text value, followed by '@'-character and the ISO 639 language code: example@EN.
- The [valueType] is only given for Properties.

Annex B. Changes to the submodel template

General

This annex lists the changes from version to version of the Submodel, together with major changes in the overall document.

Changes

This is the first version

Known issues

None

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Additional element tables

The following Submodel Element Collections form part of this Submodel template. Their attribute tables are generated directly from the Submodel template artifact, so they cannot diverge from it.

Properties of SISDevice

idShort:	SISDevice
Class:	Submodel
semanticId:	https://admin-shell.io/idta/SubmodelTemplate/SISDevice/1/0
Parent:	-
Explanation:	The submodel SIS device is a collection of properties about the functional safety requirements and information for the lifecycle of a device of a safety instrumented system (SIS) realising one or more safety instrumented functions.
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] TagInformation	https://admin-shell.io/idta/cds/TagInformation/1 gives relevant information about the tag that the device represents	[]	One
[SMC] SRSRequirements	https://admin-shell.io/idta/cds/SRSRequirements/1 SRS requirement(s) according to IEC 61511	[]	OneToMany
[SMC] DeviceModelSpecification	https://admin-shell.io/idta/cds/DeviceModelSpecification/1 gives information about the model of the device	[]	One
[SMC] ReliabilityDataDesign	https://admin-shell.io/idta/cds/ReliabilityDataDesign/1 reliability data parameter(s) with design data value(s)	[]	One
[SMC] ReliabilityInfluencingProperties	https://admin-shell.io/idta/cds/ReliabilityInfluencingProperties/1 reliability influencing property for the device	[]	ZeroToOne
[SMC] DeviceInstanceSpecification	https://admin-shell.io/idta/cds/DeviceInstanceSpecification/1 gives information about the instance of the device	[]	One
[SMC] BypassData	https://admin-shell.io/idta/cds/BypassData/1 gives information about occurred bypass(es)	[]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] Diagnostics	https://admin-shell.io/idta/cds/Diagnostics/1 gives information about the device diagnostics	[]	ZeroToMany
[SMC] DeviceOperationData	https://admin-shell.io/idta/cds/DeviceOperationData/1 reliability data parameter(s) with operational data value(s)	[]	ZeroToOne
[SMC] DemandData	https://admin-shell.io/idta/cds/DemandData/1 gives information about occurred demand(s)	[]	ZeroToOne
[SMC] FailureData	https://admin-shell.io/idta/cds/FailureData/1 gives relevant information about the occurred failure(s)	[]	ZeroToOne
[SMC] ReliabilityDataOperational	https://admin-shell.io/idta/cds/ReliabilityDataOperational/1 reliability data parameter(s) with operational data value(s)	[]	ZeroToOne
[SMC] ProofTestData	https://admin-shell.io/idta/cds/ProofTestData/1 gives information about test(s)	[]	ZeroToMany

Properties of SRSRequirements/ProofTest

idShort:	ProofTest
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ProofTest/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TestCoverage	https://admin-shell.io/idta/cds/TestCoverage/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One
[Prop] MinimumTestIntervalOperatorSpecification	https://admin-shell.io/idta/cds/MinimumTestIntervalOperatorSpecification/1 SRS requirement(s) according to IEC 61511	[xs:integer]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Maximum TestIntervalForSILCompliance	https://admin-shell.io/idta/cds/MaximumTestIntervalForSILCompliance/1 SRS requirement(s) according to IEC 61511	[xs:integer]	One
[File] TestProcedure	https://admin-shell.io/idta/cds/TestProcedure/1 SRS requirement(s) according to IEC 61511	[]	ZeroToMany
[Prop] RequirementsForTesting	https://admin-shell.io/idta/cds/RequirementsForTesting/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[SMC] FailCriterion	https://admin-shell.io/idta/cds/FailCriterion/1 SRS requirement(s) according to IEC 61511	[]	OneToMany

Properties of SRSRequirements/ProofTest/FailCriterion

idShort:	FailCriterion
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FailCriterion/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] FailCriterionFailureMode	https://admin-shell.io/idta/cds/FailCriterionFailureMode/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Prop] BooleanFailCriterionDescription	https://admin-shell.io/idta/cds/BooleanFailCriterionDescription/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] NumericFailCriterionDescriptionOfMeasurement	https://admin-shell.io/idta/cds/NumericFailCriterionDescriptionOfMeasurement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] NumericFailCriterionValue	https://admin-shell.io/idta/cds/NumericFailCriterionValue/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	ZeroToOne
[Prop] UnitOfMeasure	https://admin-shell.io/idta/cds/UnitOfMeasure/1 gives information about a measurement	[xs:string]	ZeroToOne
[Prop] NumericFailCriterionOperator	https://admin-shell.io/idta/cds/NumericFailCriterionOperator/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] CommentsToFailCriterion	https://admin-shell.io/idta/cds/CommentsToFailCriterion/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany

Properties of SRSRequirements/PartialStrokeTest

idShort:	PartialStrokeTest
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/PartialStrokeTest/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TestCoverage	https://admin-shell.io/idta/cds/TestCoverage/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One
[Prop] MinimumTestIntervalOperatorSpecification	https://admin-shell.io/idta/cds/MinimumTestIntervalOperatorSpecification/1 SRS requirement(s) according to IEC 61511	[xs:integer]	One
[Prop] MaximumTestIntervalForSILCompliance	https://admin-shell.io/idta/cds/MaximumTestIntervalForSILCompliance/1 SRS requirement(s) according to IEC 61511	[xs:integer]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[File] TestProcedure	https://admin-shell.io/idta/cds/TestProcedure/1 SRS requirement(s) according to IEC 61511	[]	ZeroToMany
[SMC] FailCriterion	https://admin-shell.io/idta/cds/FailCriterion/1 SRS requirement(s) according to IEC 61511	[]	OneToMany

Properties of SRSRequirements/PartialStrokeTest/FailCriterion

idShort:	FailCriterion
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FailCriterion/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] FailCriterionFailureMode	https://admin-shell.io/idta/cds/FailCriterionFailureMode/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Prop] BooleanFailCriterionDescription	https://admin-shell.io/idta/cds/BooleanFailCriterionDescription/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] NumericFailCriterionDescriptionOfMeasurement	https://admin-shell.io/idta/cds/NumericFailCriterionDescriptionOfMeasurement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] NumericFailCriterionValue	https://admin-shell.io/idta/cds/NumericFailCriterionValue/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	ZeroToOne
[Prop] UnitOfMeasure	https://admin-shell.io/idta/cds/UnitOfMeasure/1 gives information about a measurement	[xs:string]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] NumericFailCriterionComparison	https://admin-shell.io/idta/SISDevice/NumericFailCriterionOperator/1/0 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] CommentsToFailCriterion	https://admin-shell.io/idta/cds/CommentsToFailCriterion/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany

Properties of SRSRequirements/BypassRequirements

idShort:	BypassRequirements
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/BypassRequirements/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MaximumAllowableBypassTime	https://admin-shell.io/idta/cds/MaximumAllowableBypassTime/1 SRS requirement(s) according to IEC 61511	[xs:float]	One
[Prop] BypassAdministrativeControlsRequired	https://admin-shell.io/idta/cds/BypassAdministrativeControlsRequired/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	One
[Prop] ReferenceToBypassGroup	https://admin-shell.io/idta/cds/ReferenceToBypassGroup/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany

Properties of SRSRequirements/EnvironmentalExtremes

idShort:	EnvironmentalExtremes
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Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/EnvironmentalExtremes/1
Parent:	-
Explanation:	environmental extreme value
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MinimumAllowableTemperature	https://admin-shell.io/idta/cds/MinimumAllowableTemperature/1 environmental extreme value	[xs:integer]	ZeroToOne
[Prop] MaximumAllowableTemperature	https://admin-shell.io/idta/cds/MaximumAllowableTemperature/1 environmental extreme value	[xs:integer]	ZeroToOne
[Prop] MinimumAllowableHumidity	https://admin-shell.io/idta/cds/MinimumAllowableHumidity/1 SRS requirement(s) according to IEC 61511	[xs:integer]	ZeroToOne
[Prop] MaximumAllowableHumidity	https://admin-shell.io/idta/cds/MaximumAllowableHumidity/1 SRS requirement(s) according to IEC 61511	[xs:integer]	ZeroToOne
[Prop] ContaminantRequirement	https://admin-shell.io/idta/cds/ContaminantRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] GroundingRequirement	https://admin-shell.io/idta/cds/GroundingRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] MaximumElectromagneticInterference	https://admin-shell.io/idta/cds/MaximumElectromagneticInterference/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	ZeroToOne
[Prop] ShockResistance	https://admin-shell.io/idta/cds/ShockResistance/1 environmental extreme value	[xs:string]	ZeroToOne
[Prop] VibrationResistance	https://admin-shell.io/idta/cds/VibrationResistance/1 environmental extreme value	[xs:string]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] ElectrostaticDischargeProtection	https://admin-shell.io/idta/cds/ElectrostaticDischargeProtection/1 environmental extreme value	[xs:string]	ZeroToMany
[Prop] ATEXZone	https://admin-shell.io/idta/cds/ATEXZone/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] GasGroup	https://admin-shell.io/idta/cds/GasGroup/1 environmental extreme value	[xs:string]	ZeroToOne
[Prop] TemperatureClass	https://admin-shell.io/idta/cds/TemperatureClass/1 environmental extreme value	[xs:string]	ZeroToOne
[Prop] FloodingRequirement	https://admin-shell.io/idta/cds/FloodingRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] LightningExposureRequirement	https://admin-shell.io/idta/cds/LightningExposureRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] OtherEnvironmentalRequirement	https://admin-shell.io/idta/cds/OtherEnvironmentalRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany

Properties of SRSRequirements/FailureModeResponses

idShort:	FailureModeResponses
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FailureModeResponses/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] FailureModeAbbreviation	https://admin-shell.io/idta/cds/FailureModeAbbreviation/1 SRS requirement(s) according to IEC 61511	[xs:string]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] FailureModeName	https://admin-shell.io/idta/cds/FailureModeName/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[SMC] SafeAutomaticResponse	https://admin-shell.io/idta/cds/SafeAutomaticResponse/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne
[SMC] ManualResponse	https://admin-shell.io/idta/cds/ManualResponse/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne

Properties of SRSRequirements/FailureModeResponses/SafeAutomaticResponse

idShort:	SafeAutomaticResponse
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/SafeAutomaticResponse/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] FailurePhilosophy	https://admin-shell.io/idta/cds/FailurePhilosophy/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Ref] ReferenceToAlarm	https://admin-shell.io/idta/cds/ReferenceToAlarm/1 SRS requirement(s) according to IEC 61511	[]	ZeroToMany

Properties of SRSRequirements/FailureModeResponses/ManualResponse

idShort:	ManualResponse
Class:	SubmodelElementCollection

semanticId:	https://admin-shell.io/idta/cds/ManualResponse/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] ManualResponseActionDescription	https://admin-shell.io/idta/cds/ManualResponseActionDescription/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[File] Document	https://admin-shell.io/idta/cds/Document/1 gives information about relevant document(s)	[]	ZeroToMany

Properties of SRSRequirements/InputDevice

idShort:	InputDevice
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/InputDevice/1
Parent:	-
Explanation:	gives information about the input device subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] ProcessMeasurement	https://admin-shell.io/idta/cds/ProcessMeasurement/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne
[SMC] TripPoint	https://admin-shell.io/idta/cds/TripPoint/1 gives information about the device diagnostics	[]	ZeroToMany
[SMC] DiagnosticAlarm	https://admin-shell.io/idta/cds/DiagnosticAlarm/1 SRS requirement(s) according to IEC 61511	[]	ZeroToMany
[Prop] AlarmResetAfterShutdownIsRequired	https://admin-shell.io/idta/cds/AlarmResetAfterShutdownIsRequired/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MeasurementComparisonIsRequired	https://admin-shell.io/idta/cds/M MeasurementComparisonIsRequired/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	ZeroToOne

Properties of SRSRequirements/InputDevice/ProcessMeasurement

idShort:	ProcessMeasurement
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ProcessMeasurement/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] UnitOfMeasure	https://admin-shell.io/idta/cds/UnitOfMeasure/1 gives information about the device diagnostics	[xs:string]	One
[Prop] RangeMin	https://admin-shell.io/idta/cds/RangeMin/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One
[Prop] RangeMax	https://admin-shell.io/idta/cds/RangeMax/1 SRS requirement(s) according to IEC 61511	[xs:nonNegativeInteger]	One
[Prop] Accuracy	https://admin-shell.io/idta/cds/Accuracy/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One
[Prop] SubstituteValue	https://admin-shell.io/idta/SISDevice/SubstituteValue/1/0 SRS requirement(s) according to IEC 61511	[xs:decimal]	One

Properties of SRSRequirements/InputDevice/TripPoint

idShort:	TripPoint
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/TripPoint/1
Parent:	-

Explanation:	gives information about the device diagnostics
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TripPointName	https://admin-shell.io/idta/cds/TripPointName/1 gives information about the device diagnostics	[xs:string]	One
[Prop] TripPointValue	https://admin-shell.io/idta/cds/TripPointValue/1 gives information about the device diagnostics	[xs:decimal]	One
[Prop] HysteresisOfTripPoint	https://admin-shell.io/idta/cds/HysteresisOfTripPoint/1 gives information about the device diagnostics	[xs:decimal]	ZeroToOne
[Prop] TripDeadTime	https://admin-shell.io/idta/cds/TripDeadTime/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	ZeroToOne
[Ref] ReferenceToAlarm	https://admin-shell.io/idta/cds/ReferenceToAlarm/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne

Properties of SRSRequirements/InputDevice/DiagnosticAlarm

idShort:	DiagnosticAlarm
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/DiagnosticAlarm/1
Parent:	-
Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] AlarmName	https://admin-shell.io/idta/cds/AlarmName/1 SRS requirement(s) according to IEC 61511	[xs:string]	One
[Prop] AlarmValue	https://admin-shell.io/idta/cds/AlarmValue/1 SRS requirement(s) according to IEC 61511	[xs:decimal]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] AlarmType	https://admin-shell.io/idta/cds/AlarmType/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] OtherAlarmType	https://admin-shell.io/idta/cds/OtherAlarmType/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] AlarmMessage	https://admin-shell.io/idta/cds/AlarmMessage/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne

Properties of SRSRequirements/LogicSolver

idShort:	LogicSolver
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/LogicSolver/1
Parent:	-
Explanation:	gives information about the logic solver subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] LogicSolverResetAfterShutdownIsRequired	https://admin-shell.io/idta/cds/LogicSolverResetAfterShutdownIsRequired/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	ZeroToOne

Properties of SRSRequirements/FinalElement

idShort:	FinalElement
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FinalElement/1
Parent:	-
Explanation:	gives information about the final element subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Maximum Allowable LeakageRate	https://admin-shell.io/idta/cds/MaximumAllowableLeakageRate/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] TightShutOffIsRequired	https://admin-shell.io/idta/cds/TightShutOffIsRequired/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	ZeroToOne
[Prop] ManualOperationIsPossible	https://admin-shell.io/idta/cds/ManualOperationIsPossible/1 SRS requirement(s) according to IEC 61511	[xs:boolean]	One
[Prop] ResetAfterShutdownRequirement	https://admin-shell.io/idta/cds/ResetAfterShutdownRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToOne
[Prop] FinalElementFunction	https://admin-shell.io/idta/cds/FinalElementFunction/1 gives input to the description of the SIS device	[xs:string]	ZeroToOne

Properties of DeviceModelSpecification/FinalElement

idShort:	FinalElement
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FinalElement/1
Parent:	-
Explanation:	gives information about the final element subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] EnergizeSource	https://admin-shell.io/idta/cds/EnergizeSource/1 gives information about the model of the device	[xs:string]	ZeroToOne

Properties of ReliabilityInfluencingProperties/InputDevice

idShort:	InputDevice
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/InputDevice/1
Parent:	-
Explanation:	gives information about the input device subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Measuring Principle	https://admin-shell.io/idta/cds/MeasuringPrinciple/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] InterfaceMeasurement	https://admin-shell.io/idta/cds/InterfaceMeasurement/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] Function	https://admin-shell.io/idta/cds/Function/1 reliability influencing property for the device	[xs:string]	ZeroToOne

Properties of ReliabilityInfluencingProperties/LogicSolver

idShort:	LogicSolver
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/LogicSolver/1
Parent:	-
Explanation:	gives information about the logic solver subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] DesignPrinciple	https://admin-shell.io/idta/cds/DesignPrinciple/1 reliability influencing property for the device	[xs:string]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] DeviceType	https://admin-shell.io/idta/cds/DeviceType/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] Application	https://admin-shell.io/idta/cds/Application/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] StyleOfFailSafe	https://admin-shell.io/idta/cds/StyleOfFailSafe/1 reliability influencing property for the device	[xs:string]	ZeroToOne

Properties of ReliabilityInfluencingProperties/FinalElement

idShort:	FinalElement
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FinalElement/1
Parent:	-
Explanation:	gives information about the final element subsystem of the SIF
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] DeviceType	https://admin-shell.io/idta/cds/DeviceType/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] DesignPrinciple	https://admin-shell.io/idta/cds/DesignPrinciple/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] ActuationPrinciple	https://admin-shell.io/idta/cds/ActuationPrinciple/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] ValveSize	https://admin-shell.io/idta/cds/ValveSize/1 reliability influencing property for the device	[xs:decimal]	ZeroToOne
[Prop] DamperLength	https://admin-shell.io/idta/cds/DamperLength/1 reliability influencing property for the device	[xs:decimal]	ZeroToOne
[Prop] DamperWidth	https://admin-shell.io/idta/cds/DamperWidth/1 reliability influencing property for the device	[xs:decimal]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] MountingPrinciple	https://admin-shell.io/idta/cds/MountingPrinciple/1 reliability influencing property for the device	[xs:string]	ZeroToOne
[Prop] DoorHasPersonnelProtectionSystem	https://admin-shell.io/idta/cds/DoorHasPersonnelProtectionSystem/1 reliability influencing property for the device	[xs:boolean]	ZeroToOne
[Prop] VoltageClass	https://admin-shell.io/idta/cds/VoltageClass/1 reliability influencing property for the device	[xs:string]	ZeroToOne

Properties of BypassData/BypassEvent

idShort:	BypassEvent
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/BypassEvent/1
Parent:	-
Explanation:	gives information about occurred bypass(es)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TimeStampDateAndTime	0112/2///61987#ABN627#001 date of the event given within the current SMC	[xs:time]	One
[Prop] BypassDescription	https://admin-shell.io/idta/cds/BypassDescription/1 gives information about occurred bypass(es)	[xs:string]	ZeroToOne
[Prop] BypassDuration	https://admin-shell.io/idta/cds/BypassDuration/1 gives information about occurred bypass(es)	[xs:integer]	One

Properties of Diagnostics/AlarmEventData

idShort:	AlarmEventData
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/AlarmEventData/1

Parent:	-
Explanation:	gives information about the device diagnostics
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] AlarmDescription	https://admin-shell.io/idta/cds/AlarmDescription/1 gives information about the device diagnostics	[xs:string]	ZeroToOne
[Prop] TimeStampDateAndTime	0112/2///61987#ABN627#001 date of the event given within the current SMC	[xs:time]	One
[Prop] AlarmSource	https://admin-shell.io/idta/cds/AlarmSource/1 gives information about the device diagnostics	[xs:string]	One
[Prop] AlarmState	https://admin-shell.io/idta/cds/AlarmState/1 gives information about the device diagnostics	[xs:string]	ZeroToOne
[Prop] AlarmStatus	https://admin-shell.io/idta/cds/AlarmStatus/1 gives information about the device diagnostics	[xs:string]	ZeroToOne
[Ref] ReferenceToFailureEvent	https://admin-shell.io/idta/cds/ReferenceToFailureEvent/1 gives information about occurred test(s) or alarm(s)	[]	ZeroToOne

Properties of DemandData/DeviceDemandEvent

idShort:	DeviceDemandEvent
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/DeviceDemandEvent/1
Parent:	-
Explanation:	gives information about occurred demand(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TimeStampDateAndTime	0112/2///61987#ABN627#001 date of the event given within the current SMC	[xs:time]	One
[Prop] DemandDescription	https://admin-shell.io/idta/cds/DemandDescription/1 gives information about occurred demand(s)	[xs:string]	ZeroToOne
[Prop] DemandClass	https://admin-shell.io/idta/cds/DemandClass/1 gives information about occurred demand(s)	[xs:string]	ZeroToOne
[Prop] SISDeviceResponseTimeActual	https://admin-shell.io/idta/cds/SISDeviceResponseTimeActual/1 gives information about occurred demand(s)	[xs:decimal]	ZeroToOne
[Ref] ReferenceToFailureEvent	https://admin-shell.io/idta/cds/ReferenceToFailureEvent/1 gives information about occurred test(s) or alarm(s)	[]	ZeroToOne
[Prop] IsSIFDemand	https://admin-shell.io/idta/cds/IsSIFDemand/1 gives information about occurred demand(s)	[xs:boolean]	One

Properties of FailureData/FailureEvent

idShort:	FailureEvent
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/FailureEvent/1
Parent:	-
Explanation:	gives information about occurred failure(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TimeStampDateAndTime	0112/2///61987#ABN627#001 date of the event given within the current SMC	[xs:time]	One
[Prop] DetectionMethod	https://admin-shell.io/idta/cds/DetectionMethod/1 gives information about occurred failure(s)	[xs:string]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Detection Mode	https://admin-shell.io/idta/cds/DetectionMode/1 gives information about occurred failure(s)	[xs:string]	One
[Prop] FailureMode	https://admin-shell.io/idta/cds/FailureMode/1 gives information about occurred failure(s)	[xs:string]	ZeroToOne
[Prop] FailureClass	https://admin-shell.io/idta/cds/FailureClass/1 gives information about occurred failure(s)	[xs:string]	One
[Prop] IsSpurious Trip	https://admin-shell.io/idta/cds/IsSpuriousTrip/1 gives information about occurred failure(s)	[xs:boolean]	One
[Prop] FailureMechanism	https://admin-shell.io/idta/cds/FailureMechanism/1 gives information about occurred failure(s)	[xs:string]	ZeroToMany
[Prop] IsSystematicOrRandomHardwareFailure	https://admin-shell.io/idta/cds/IsSystematicOrRandomHardwareFailure/1 gives information about occurred failure(s)	[xs:string]	ZeroToOne
[Prop] FailureDescription	https://admin-shell.io/idta/cds/FailureDescription/1 gives information about occurred failure(s)	[xs:string]	OneToMany
[Prop] FailureCause	https://admin-shell.io/idta/cds/FailureCause/1 gives information about occurred failure(s)	[xs:string]	ZeroToOne
[Prop] FailureIndependencyType	https://admin-shell.io/idta/cds/FailureIndependencyType/1 gives information about occurred failure(s)	[xs:string]	ZeroToOne
[Prop] IsSIFFailure	https://admin-shell.io/idta/cds/IsSIFFailure/1 gives information about occurred failure(s)	[xs:boolean]	ZeroToOne
[Prop] ExpectedRepairTime	https://admin-shell.io/idta/cds/ExpectedRepairTime/1 gives information about occurred failure(s)	[xs:double]	ZeroToOne
[Prop] ActualRepairTime	https://admin-shell.io/idta/cds/ActualRepairTime/1 gives information about occurred failure(s)	[xs:double]	One
[Prop] NotificationNumber	https://admin-shell.io/idta/cds/NotificationNumber/1 gives information about occurred failure(s)	[xs:integer]	One

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] WorkOrderNumber	https://admin-shell.io/idta/cds/WorkOrderNumber/1 gives information about occurred failure(s)	[xs:integer]	ZeroToOne
[Prop] Measure	https://admin-shell.io/idta/cds/Measure/1 gives information about occurred failure(s)	[xs:string]	ZeroToMany

Properties of ProofTestData/ProofTestEvent

idShort:	ProofTestEvent
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/ProofTestEvent/1
Parent:	-
Explanation:	gives information about test(s)
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] TimeStampDateAndTime	0112/2///61987#ABN627#001 date of the event given within the current SMC	[xs:time]	One
[Prop] TestLocation	https://admin-shell.io/idta/cds/TestLocation/1 gives information about test(s)	[xs:string]	ZeroToOne
[Prop] TestResultFailedOrPassed	https://admin-shell.io/idta/cds/TestResultFailedOrPassed/1 gives information about test(s)	[xs:string]	One
[Prop] TestResultNumeric	https://admin-shell.io/idta/cds/TestResultNumeric/1 gives information about test(s)	[xs:float]	ZeroToOne
[Prop] TestResultText	https://admin-shell.io/idta/cds/TestResultText/1 gives information about test(s)	[xs:string]	ZeroToOne
[Prop] CommentToTestResult	https://admin-shell.io/idta/cds/CommentToTestResult/1 gives information about test(s)	[xs:string]	ZeroToOne

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Ref] Reference ToFailure Event	https://admin-shell.io/idta/cds/ReferenceToFailureEvent/1 gives information about occurred test(s) or alarm(s)	[]	ZeroToOne